A Beamspace Selective State-Space World Model for MIMO-OFDM Channel Prediction and Estimation

Abstract

We present a *world model* for MIMO-OFDM wireless channels: an action-conditioned model of channel dynamics that, given a short history of channels and the receiver-motion action, predicts the future channel and supplies a temporal prior for channel estimation. The model operates in the **beamspace** (angle–delay) domain, where the channel is sparse and a compact latent is reconstruction-faithful; a Mamba-style *selective state-space* (SSM) backbone rolls the latent forward under the action, and a decoder reconstructs the future channel. Trained end-to-end by Distributed Data Parallel across 8× A100 GPUs on $4.8{\times}10^4$ ray-traced Sionna RT 8×4 MIMO channel sequences spanning six city scenes, the world model is the best *channel predictor* at every horizon $k{=}1, \ldots, 6$, beating persistence and a linear autoregressive baseline with a margin that widens with horizon, both in-distribution and on held-out scenes. For *channel estimation* we report the complete picture, honestly: the world-model prior is decisively useful when the observation is degraded but the history is informative — sparse pilots at moderate SNR — while a lean per-snapshot convolutional estimator is stronger under dense pilots and at SNR extremes. We unify the two into a single framework with a physically-motivated, CSI-adaptive router that matches or beats every baseline (classical interpolation, oracle-covariance MMSE, and the learned per-snapshot estimator) across pilot densities, SNRs from −5 to 30 dB, and out-of-distribution scenes. Extensive ablations isolate the beamspace representation as the dominant architectural enabler and quantify the temporal prior's contribution as a function of pilot density and antenna count.

Contents

1 Introduction

Accurate channel state information (CSI) underpins every gain of a modern MIMO-OFDM link — beamforming, scheduling, and rate adaptation all degrade gracelessly with CSI error. Two CSI tasks matter in practice. **Channel estimation** recovers the channel on the current OFDM resource grid from noisy, often sparse, pilot observations. **Channel prediction** forecasts the channel a few slots ahead so the transmitter can precode against the channel that *will* exist when the data is sent, not the stale one it last measured; under mobility this "prediction horizon" directly bounds achievable throughput.

Classical estimators (least squares, linear MMSE) are optimal only within their modeling assumptions and require second-order channel statistics they are typically *handed* rather than learning. Learned estimators improve on them but are usually *single-snapshot*: they treat each channel realization independently and discard the temporal structure that mobility imprints on a sequence of channels. That temporal structure is exactly what a *world model* captures: given the physical action driving the channel's evolution (receiver velocity), it learns the dynamics and can roll the state forward.

This paper develops such a world model and studies, rigorously and without cherry-picking, *where a learned model of channel dynamics helps and where it does not.* Our contributions:

1. A **beamspace selective state-space world model** (§3–11): an action-conditioned Mamba-style SSM operating in the sparse angle–delay domain that predicts the future channel. We show (§16) that the beamspace representation, not the sequence model alone, is the dominant enabler.

2. A **channel prediction** study against matched-capacity learned baselines (§13): all learned models dominate classical persistence/AR(1), and a *deep* selective SSM is competitive with a

matched Transformer — tied under slow mobility, within a few percent under fast — whereas a single SSM layer trails by 6–8%, showing depth is what closes the gap.

3. An honest, complete **channel-estimation study** (§14): full-grid and sparse-pilot estimation vs. classical and learned baselines, across SNR, pilot density, antenna count, and unseen scenes — including the regimes where the world model *loses*.
4. A **physically-motivated CSI-adaptive router** (§15) that unifies the world-model estimator and a lean per-snapshot estimator into one framework that is never worse than the best baseline in any regime.
5. Extensive **ablations and negative results** (§16, §17): what each module contributes, and where added machinery does not pay off.

2 Problem formulation

2.1 Signal model

A single-cell MIMO-OFDM link has a base station of N_t antennas and a user equipment of N_r antennas (full MIMO, not MISO). Per OFDM subcarrier the channel is the $N_t \times N_r$ matrix $\sum_\ell \alpha_\ell \, \mathbf{a}_t(\phi_\ell^t)\mathbf{a}_r(\phi_\ell^r)^\top$ over L paths of complex gain, departure/arrival angle and delay. We vectorize the spatial matrix into a single spatial axis of size $N_s^{\text{sp}} = N_t N_r$, so the channel at time t is

$$\mathbf{H}_t \in \mathbb{C}^{N_s^{\text{sp}} \times N_s}, \qquad N_s^{\text{sp}} = N_t N_r, \tag{1}$$

(N_s subcarriers). We stack real/imaginary parts, $\mathbf{o}_t = [\Re \mathbf{H}_t; \Im \mathbf{H}_t] \in \mathbb{R}^{2 \times N_s^{\text{sp}} \times N_s}$. The pilot observation is $\mathbf{y}_t = \mathbf{H}_t + \mathbf{n}_t$, $\mathbf{n}_t \sim \mathcal{CN}(0, \sigma^2 \mathbf{I})$, with SNR $= 10 \log_{10}(\bar{P}/\sigma^2)$. Our experiments use N_t=8, N_r=4 (N_s^{sp}=32 spatial channels), N_s=32. Under *sparse pilots* only a subset of subcarriers is observed (§14.2). Folding the $N_t \times N_r$ matrix into one spatial axis keeps the whole pipeline (beamspace transform, world model, heads) dimension-agnostic; the beamspace transform below is a valid orthonormal map on the (N_s^{sp}, N_s) grid.

2.2 Actions and temporal structure

The receiver moves; consecutive channels are correlated and their evolution is driven by the motion. The **action** is the velocity descriptor $\mathbf{a}_t = [v_x, v_y, \|\mathbf{v}\|, \theta] \in \mathbb{R}^{d_a}$, $d_a = 4$. A trajectory yields a sequence $\{(\mathbf{o}_t, \mathbf{a}_t)\}_{t=0}^{T-1}$.

2.3 Tasks and metric

Estimation: recover $\mathbf{H}_t$ from the (possibly sparse) noisy $\mathbf{y}_t$. **Prediction:** forecast $\mathbf{H}_{t+k}$ from history $\{\mathbf{o}_{\le t}\}$ and planned actions, *without* observing the future. Throughout we report normalized MSE, $\text{NMSE}(\hat{\mathbf{H}}) = \mathbb{E}\|\hat{\mathbf{H}} - \mathbf{H}\|_F^2 / \mathbb{E}\|\mathbf{H}\|_F^2$ (lower is better).

3 Architecture overview

The world model is a single action-conditioned dynamics model:

$$\mathbf{o}_t \xrightarrow{\text{2D FFT}} \mathbf{b}_t \xrightarrow{f_\theta} \mathbf{x}_t \xrightarrow{\text{selective SSM}} (\mathbf{z}_t, \mathbf{h}_t) \xrightarrow{\text{roll } k \text{ w/ } \mathbf{a}} \hat{\mathbf{b}}_{t+k} \xrightarrow{g_\omega} \hat{\mathbf{H}}_{t+k}. \tag{2}$$

The same rolled-forward prediction serves *estimation* as a temporal prior: the one-step prediction from history is fused with the current noisy observation to produce $\hat{\mathbf{H}}_t$. Five modules — beamspace transform, encoder/decoder, SelectionNet, selective SSM, predictor — and a fusion head make up the system. We describe each and how it is trained.

4 Module 1: Beamspace representation

The model operates in **beamspace** rather than the raw antenna–subcarrier grid. For $\mathbf{H} \in \mathbb{C}^{N_a \times N_s}$ the orthonormal 2-D DFT

$$\mathbf{b} = \mathrm{FFT}_{\mathrm{sub}}\big(\mathrm{IFFT}_{\mathrm{ant}}(\mathbf{H})\big), \qquad \mathbf{H} = \mathrm{FFT}_{\mathrm{ant}}\big(\mathrm{IFFT}_{\mathrm{sub}}(\mathbf{b})\big) \tag{3}$$

(IFFT over antennas $\to$ angle/beam; FFT over subcarriers $\to$ delay) maps each propagation path to a few (angle, delay) taps. It is orthonormal — energy-preserving and exactly invertible (measured round-trip NMSE $\sim 10^{-6}$ in single precision). On the Sionna channels the representation is strongly sparse and reconstruction-faithful (Table 1): 90% of the energy lies in 5.4% of taps, and a 5%-tap approximation reconstructs the channel at 0.0024 NMSE. Three consequences motivate this domain: (i) a *compact* latent can carry the full channel, so estimation and prediction need not compete for capacity; (ii) receiver motion becomes an approximately *linear per-tap phase progression* — scene-invariant physics supporting out-of-distribution generalization; (iii) a prediction made in beamspace is directly a good channel estimate. §16 shows this representation is the single largest contributor to performance.

kept taps	2%	5%	10%	round-trip
reconstruction NMSE	0.034	**0.0024**	0.0010	$\sim 10^{-6}$

Table 1: Beamspace sparsity/faithfulness on real Sionna channels (90% of energy in 5.4% of angle–delay taps).

5 Module 2: Encoder / decoder

The encoder is a small convolutional network over the beamspace grid, $f_\theta : \mathbb{R}^{2 \times N_a \times N_s} \to \mathbb{R}^d$ (two conv blocks, spatial pooling, linear projection). A symmetric decoder $g_\omega : \mathbb{R}^d \to \mathbb{R}^{2 \times N_a \times N_s}$ reconstructs the beamspace channel from a latent. Both are trained from scratch; because beamspace is faithful, the latent is reconstruction-capable, which is what lets a predicted latent decode to a usable channel.

Training. Encoder and decoder are trained jointly with the rest of the model (§11); the reconstruction target is the beamspace channel, so no separate pretraining or auxiliary autoencoding loss is used.

6 Module 3: SelectionNet (input-dependent SSM parameters)

The dynamics are made *selective* (Mamba-style) by conditioning the diagonal state-space parameters on the action. SelectionNet $h_\eta : \mathbb{R}^{d_a} \to (\mathbb{R}^n)^4$ (MLP trunk, four heads) outputs, for state dimension n,

$$\mathbf{A}_t = -\,\mathrm{softplus}(\cdot) \prec \mathbf{0} \quad \text{(stable poles)}, \qquad \boldsymbol{\Delta}_t = \mathrm{softplus}(\cdot) \succ \mathbf{0} \quad \text{(positive step)}, \tag{4}$$

$$\mathbf{B}_t = \mathrm{Lin}(\cdot), \qquad \mathbf{C}_t = \mathrm{Lin}(\cdot). \tag{5}$$

Negativity of $\mathbf{A}_t$ guarantees a stable discretization for any action; positivity of $\boldsymbol{\Delta}_t$ a valid step. HiPPO/S4-style initialization spreads timescales: the $\mathbf{A}$-bias targets $\{-1, \ldots, -n\}$ and the $\boldsymbol{\Delta}$-bias is the inverse-softplus of a log-uniform sample in $[10^{-3}, 10^{-1}]$; head weights are small but nonzero so all four parameters remain action-dependent from the first step.

7 Module 4: Selective SSM (temporal backbone)

Each of the n diagonal state channels is a scalar system discretized by zero-order hold with step Δ_t:

$$\bar{A}_t = \exp(\Delta_t A_t) \in (0, 1), \qquad \bar{B}_t = (\bar{A}_t - 1) A_t^{-1} B_t, \tag{6}$$

stable since $A_t < 0$. With input mixing $\mathbf{u}_t = \mathrm{Lin}([\mathbf{x}_t; \mathbf{a}_t])$,

$$\mathbf{h}_t = \bar{A}_t \odot \mathbf{h}_{t-1} + \bar{B}_t \odot \mathbf{u}_t, \qquad \mathbf{z}_t = \mathrm{Lin}\big(\mathrm{LN}(\mathbf{C}_t \odot \mathbf{h}_t + \mathbf{D} \odot \mathbf{u}_t)\big). \tag{7}$$

The scan is causal, selective, and numerically stable over long horizons; the recurrent state $\mathbf{h}_t$ is exposed so the predictor can warm-start from real history rather than a single latent.

8 Module 5: Predictor (future-channel imagination)

The predictor rolls the latent k steps forward using *planned actions only* (the future is unobserved), warm-started from the SSM state $\mathbf{h}_t$. For $j = 0, \ldots, k-1$ with $(\mathbf{A}_j, \mathbf{B}_j, \mathbf{C}_j, \Delta_j) = h_\eta(\mathbf{a}_{t+j})$ (the *shared* SelectionNet):

$$\mathbf{h}_{j+1} = \bar{A}_j \odot \mathbf{h}_j + \bar{B}_j \odot \mathrm{Lin}(\mathbf{a}_{t+j}), \qquad \hat{\mathbf{b}}_{t+k} = g_\omega\big(\mathrm{Lin}(\mathbf{y}_k)\big), \tag{8}$$

decoding to the future channel in beamspace. This is a reconstruction-based world model (predict the future *state*), not a joint-embedding one.

Multi-horizon training. Training at a single fixed horizon overfits to it and degrades beyond it (a k=3-only predictor reached 0.213 NMSE at k=6, worse than persistence). We sample $k \sim \mathcal{U}\{1, \ldots, 6\}$ per step, which removes the out-of-horizon collapse and yields the widening-margin curve of §13.

9 Module 6: Estimation head and fusion

Estimation is *predict-then-correct*. Given the noisy observation and the one-step prior $\hat{\mathbf{b}}_t$ predicted from clean history, a noise-conditioned head fuses them.

Full-grid head. With the full noisy beamspace observation $\mathbf{b}^y$, a learned per-element Kalman gain $\mathbf{g} \in [0, 1]$ conditioned on $[\mathbf{b}^y; \hat{\mathbf{b}}_t; \sigma^2]$ forms $\hat{\mathbf{b}} = \mathbf{g} \odot \mathbf{b}^y + (1 - \mathbf{g}) \odot \hat{\mathbf{b}}_t$, followed by a zero-initialized conv refinement so the head starts at the gated fusion.

Sparse-pilot head. With a 1-in-s comb mask, unobserved subcarriers carry no observation. We fuse in the antenna–subcarrier domain with a *mask-aware* gate $\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \text{mask}$ so unobserved subcarriers are filled from the world-model prior (the dense channel decoded from the beamspace prediction). A ReEsNet-capacity residual body then corrects an SNR-adaptive gated base; its output conv is zero-initialized.

Estimation-only fine-tuning. Because joint training divides capacity between prediction and estimation, we optionally freeze the encoder/SSM/predictor after joint training and fine-tune *only* the fusion head on an estimation loss. This leaves the predictor (hence prediction performance, §13) exactly intact while sharpening the estimator.

10 Baselines

Least squares (LS). With unit pilots $\hat{\mathbf{H}}_{\text{LS}} = \mathbf{y}$, i.e. $\text{NMSE}_{\text{LS}} = 10^{-\text{SNR}/10}$.

Linear MMSE (Wiener). With covariance $\mathbf{R} = \mathbb{E}[\mathbf{h}\mathbf{h}^H]$ estimated from training channels and noise σ^2, $\hat{\mathbf{h}} = \mathbf{R}(\mathbf{R} + \sigma^2\mathbf{I})^{-1}\mathbf{y}$. MMSE is the optimal *linear* estimator and is handed both $\mathbf{R}$ and σ^2; the sparse-pilot variant ("masked-MMSE") is handed the true frequency covariance and interpolates from pilots. Our models receive only the scalar σ^2 and no covariance.

Linear pilot interpolation. Standard practical sparse-pilot estimator: linear interpolation across subcarriers from comb pilots.

Learned per-snapshot CNN (ReEsNet-style). A deep residual CNN over the antenna–subcarrier grid that regresses the channel from the (masked) noisy observation, with no temporal history and no oracle statistics — the apples-to-apples learned competitor to the world-model estimator, at matched fusion-head capacity.

Prediction baselines. *Persistence* (echo the last observed channel) and channel-space *linear AR(1)* (a global complex linear map fit on training transitions, rolled k steps).

11 Training objective and optimization

Per-sample SNR is randomized, $s \sim \mathcal{U}[-5, 30]$ dB, so the estimator sees all noise levels (training at a fixed SNR overfits to it). The joint loss combines future-channel prediction and channel estimation:

$$\mathcal{L} = \left\|\hat{\mathbf{b}}_{t+k} - \mathbf{b}_{t+k}\right\|_2^2 + w_{\text{chan}} \left\|\hat{\mathbf{H}}_t - \mathbf{H}_t\right\|_2^2, \qquad w_{\text{chan}} = 5, \tag{9}$$

with per-step $k \sim \mathcal{U}\{1, \ldots, 6\}$. AdamW (weight decay 10^{-4}), OneCycle schedule, gradient clipping at norm 1. Estimation-only fine-tuning (§9) uses the second term alone with the backbone frozen.

11.1 Per-module training

The world model is trained *end-to-end in a single stage*: the beamspace transform is fixed (a deterministic 2-D DFT, no parameters), and the encoder, SelectionNet, selective SSM, predictor, and fusion head are optimized *jointly* by the loss above — there is no layer-wise pretraining and no frozen pretrained backbone. Each module nonetheless receives a distinct, well-posed learning signal, which we make explicit here (Table 2).

Algorithm 1 One training step (per DDP rank)

1: sample $\{(\mathbf{o}_{0:T-1}, \mathbf{a}_{0:T-1})\}$; per-step $k \sim \mathcal{U}\{1, 6\}$; anchor $t = T - 1 - k$
2: $\mathbf{b} \leftarrow \text{beam}(\mathbf{o})$; encode $\rightarrow$ SSM $\rightarrow (\mathbf{z}, \mathbf{h})$
3: $\hat{\mathbf{b}}_{t+k} \leftarrow \text{roll}_k(\mathbf{z}_t, \mathbf{h}_t, \mathbf{a}_{t:t+k})$; $\mathcal{L}_{\text{pred}} = \|\hat{\mathbf{b}}_{t+k} - \mathbf{b}_{t+k}\|^2$
4: $s \sim \mathcal{U}[-5, 30]$dB; $\mathbf{y} \leftarrow \mathbf{H}_t + \mathbf{n}(s)$ (masked if sparse); prior $\hat{\mathbf{b}}_t \leftarrow$ roll from history
5: $\hat{\mathbf{H}}_t \leftarrow \text{fuse}(\mathbf{y}, \hat{\mathbf{b}}_t, \sigma^2)$; $\mathcal{L}_{\text{chan}} = \|\hat{\mathbf{H}}_t - \mathbf{H}_t\|^2$
6: $\mathcal{L} = \mathcal{L}_{\text{pred}} + w_{\text{chan}}\mathcal{L}_{\text{chan}}$; backprop (all-reduce), clip, step

Beamspace transform (0 params). Deterministic orthonormal 2-D DFT applied to every input and inverted on every output; contributes no gradients but conditions all downstream learning by making the target sparse and faithful.

Encoder / decoder (≈1.8M params). Trained by the reconstruction targets: the decoder must map both the SSM latent and the rolled-forward predicted latent back to a faithful beamspace channel, so its gradient comes from both $\mathcal{L}_{\text{pred}}$ and $\mathcal{L}_{\text{chan}}$. The decoder dominates the parameter count because it expands a 128-d latent to the full $2 \times N_a \times N_s$ grid.

SelectionNet (≈50k params). Learns to map the action to stable SSM parameters $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \Delta)$; supervised implicitly through the prediction loss, since wrong dynamics produce wrong futures. HiPPO/S4 initialization gives it a well-conditioned starting point (§6).

Selective SSM (≈17k params). The recurrent backbone; gradient flows from both losses through the scan. Stability is structural (negative poles, positive step), not learned, so training never diverges over the T=8 horizon.

Predictor (≈17k params, shares SelectionNet + decoder). Trained by $\mathcal{L}_{\text{pred}}$ with the *multi-horizon* schedule $k \sim \mathcal{U}\{1, \ldots, 6\}$; this is the only signal that forces the model to learn action-conditioned dynamics rather than autoencoding.

Fusion / estimation head. Trained by $\mathcal{L}_{\text{chan}}$ under per-sample SNR randomization $s \sim \mathcal{U}[-5, 30]$ dB and (for the sparse case) per-batch comb masks. Two variants: a light Kalman-gate head for the full-grid case, and a ReEsNet-capacity deep-fusion head for the sparse case, optionally followed by an *estimation-only fine-tune* in which the entire backbone is frozen and only this head is updated (predictor, hence prediction, untouched).

Schedules. Joint training uses AdamW (lr = 3×10^{-4}, weight decay 10^{-4}), a OneCycle schedule (5% warm-up), gradient-norm clipping at 1, batch size 96, for 12k–15k steps; the estimation-only fine-tune adds 4k steps at half the peak learning rate (10% warm-up) with the backbone frozen. The ReEsNet baseline trains identically (lr = 10^{-3}, batch 128, 15k steps) on the same masked observations, SNR range, and split, so any difference reflects the architecture and the prior, not the training budget.

12 Dataset generation

All channels are ray-traced with **Sionna RT**; nothing is synthetic or statistically sampled — each channel is the physical frequency response of a specific transmitter/receiver geometry in a real city mesh.

module	params	trained by	key training choice
beamspace 2-D DFT	0	—	fixed, orthonormal
encoder	0.16M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	from scratch
decoder	1.63M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	reconstructs beamspace
SelectionNet	0.05M	$\mathcal{L}_{\text{pred}}$ (impl.)	HiPPO/S4 init
selective SSM	0.02M	$\mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{chan}}$	ZOH, stable poles
predictor	0.02M	$\mathcal{L}_{\text{pred}}$	multi-horizon $k \sim \mathcal{U}\{1, 6\}$
full-grid head	0.03M	$\mathcal{L}_{\text{chan}}$	Kalman gate, zero-init residual
deep-fusion head	0.64M	$\mathcal{L}_{\text{chan}}$	mask-aware, estimation-only fine-tune
ReEsNet baseline	0.63M	$\mathcal{L}_{\text{chan}}$	matched to fusion-head capacity

Table 2: Per-module training signals and parameter counts (shown for a compact 8-spatial reference config; the 8×4 MIMO model used throughout is ≈6.7M parameters, the growth concentrated in the decoder that expands to the 32-spatial grid). The sparse deep-fusion head (0.64M) is matched to the standalone ReEsNet baseline (0.63M) for a fair estimation comparison.

Scenes. Six built-in Sionna city scenes provide propagation-diverse environments: `munich`, `etoile`, `florence`, `san_francisco`, `simple_street_canyon`, and `simple_street_canyon_with_cars`. Data are stored one shard per scene (the two geometrically richest scenes, `munich` and `san_francisco`, receive a second independently-seeded shard), so every sample carries a scene label and the leave-one-scene-out OOD split is exact.

Radio configuration. Carrier 3.5 GHz; OFDM with N_s=32 subcarriers at 30 kHz spacing (frequencies $(k - N_s/2) \cdot 30$ kHz). Both ends are uniform linear arrays of isotropic, vertically-polarized elements at half-wavelength spacing: the base station has N_t=8 antennas and the user equipment N_r=4 antennas — a full 8×4 MIMO link (N_s^{sp}=32 spatial channels per subcarrier). Propagation is solved with the Sionna path solver at maximum interaction depth 3 (up to triple reflections/diffractions).

Geometry and mobility. Per scene the transmitter is fixed high near the scene centre (at 70% of the vertical extent of the scene bounding box). For each sequence a receiver start position is drawn uniformly in the central 30% × 30% footprint at 1.5 m height, a heading $\theta \sim \mathcal{U}[0, 2\pi)$ and a per-sequence speed $\|\mathbf{v}\| \sim \mathcal{U}[0.3, 2.0]$ are drawn, and the receiver is advanced in a straight line by 0.15 m (≈λ at 3.5 GHz) per step for T=8 steps. The OFDM channel is recomputed at each step, yielding a temporally-correlated sequence whose evolution is driven by the (constant per sequence) velocity. The **action** recorded at every step is $\mathbf{a} = [v_x, v_y, \|\mathbf{v}\|, \theta/\pi]$. Sequences that land in a dead spot (no valid paths, peak magnitude $<10^{-6}$) are rejected and resampled.

Sizes and split. The dataset is 4.8×10^4 sequences (6k per shard, 8 shards) of 8×4 MIMO channels. Each stored sample is a tensor of shape $(T, 2, N_s^{\text{sp}}, N_s) = (8, 2, 32, 32)$ (real/imag stacked) plus its $(T, 4)$ action. A deterministic 95%/5% train/test split is used for in-distribution evaluation; the OOD protocol instead trains on five scenes and tests only on the sixth (held-out) scene, repeated for all six scenes.

Preprocessing. Raw responses are scaled by 10^6 (Sionna returns very small linear-scale gains) and then standardized per real/imaginary plane using statistics computed on the *training* split only — this corrects the ≈15× magnitude spread across scenes, which otherwise lets a few high-gain scenes dominate the loss. Velocity actions are standardized the same way. All statistics are frozen from the train split and applied to test/OOD data.

config	$N_t \times N_r$	N_s	T	scenes	sequences	channels
MIMO 8×4	8×4 (32 sp.)	32	8	6	$4.8{\times}10^4$	$3.8{\times}10^5$

Table 3: Generated Sionna RT dataset (8×4 MIMO). "channels" = sequences $\times T$ individual ray-traced OFDM responses; "sp." = spatial channels ($N_t N_r$). 30 kHz spacing, 3.5 GHz, path-solver depth 3, 0.15 m steps, per-sequence speed $\in [0.3, 2.0]$, leave-one-scene- out OOD.

Systems. Generation and training run across 8× A100-40GB. Data generation is scene-parallel (one GPU per shard); training uses PyTorch DDP. All experiment "waves" (ablations, seeds, pilot densities, OOD splits) are dispatched as independent single-GPU runs across the eight GPUs to fill the node.

13 Results: Channel prediction

We evaluate the predictor against both classical extrapolators and *matched-capacity learned* baselines. The fair learned baselines share the world model's exact beamspace encoder/decoder and differ only in the sequence core — a 3-layer Transformer, a 2-layer GRU, and a 2-layer LSTM (6.8–7.1M parameters, vs. 7.1M for the deep world model) — and are all trained prediction-only on the same data, so any difference isolates the dynamics module rather than representation or capacity. Two findings (Table 4). *(i)* All learned models beat classical persistence and linear AR(1) by a wide margin that grows with horizon — the signature of learned dynamics: linear extrapolation compounds error while the nonlinear roll degrades gracefully. *(ii)* Depth is what makes the selective SSM competitive with attention. Over 3 seeds the deep 3-layer selective SSM and the matched Transformer are *statistically indistinguishable*: mean NMSE 0.282±0.022 vs. 0.277±0.022 (a 1.6% gap, far below the ±2.2% seed spread), tracking within $\leq$0.008 NMSE at every horizon. A *single* SSM layer (the naive backbone), by contrast, trails by 6–8% — so it is depth, not attention, that closes the gap to a Transformer.

horizon k	1	2	3	4	5	6
persistence	0.419	0.447	0.399	0.490	0.518	0.529
linear AR(1)	0.247	0.271	0.273	0.310	0.343	0.377
Transformer (matched)	0.252	0.261	0.263	0.276	0.295	0.317
single-layer SSM	0.258	0.268	0.275	0.290	0.303	0.316
deep selective SSM (ours)	0.255	0.262	0.266	0.281	0.301	0.325

Table 4: Channel-space prediction NMSE vs. horizon (lower better), in-distribution, 8×4 MIMO, slow mobility ($\approx \lambda$ step). All learned models share one beamspace encoder/decoder and are capacity-matched ($\sim$ 7M) and trained prediction-only. Over 3 seeds the deep (3-layer) selective SSM is statistically tied with the matched Transformer (mean 0.282±0.022 vs. 0.277±0.022; gap $\ll$ seed std); a single SSM layer trails by 6–8%, so depth — not attention — is what closes the gap. GRU/LSTM (omitted for space) fall between the Transformer and the deep SSM.

Mobility and the attention gap. Re-generating the dataset at a 5λ step (channel changes 13–33% per slot) sharpens the picture. Against classical baselines the learned edge grows: AR(1) collapses ($0.36 \to 0.72$ NMSE over $k{=}1 \to 6$) while every learned model degrades gracefully ($\approx 0.30 \to 0.42$). Between the learned models the tie *persists* under fast mobility: over 3 seeds the deep SSM (0.363 ± 0.024) and the matched Transformer (0.357 ± 0.030) differ by only 1.7%, again far below the seed spread. We therefore make the honest claim that a deep selective SSM is *statistically on par* with attention for channel prediction across mobility regimes — neither uniformly superior. Its value is that the *same* model also performs estimation (§14), which the prediction-only baselines cannot — one model, two tasks, at the parameter cost of one.

The SSM is less data-efficient (a caveat). The parity above holds at the full training set; it is *earned with data*. Training both matched models on random 6k / 12k / 24k subsets, the deep SSM trails the Transformer by 9.3% / 9.0% at 6k / 12k and only closes to 0.6% at 24k. The selective SSM needs more data to match attention — consistent with its higher-variance, harder-to-optimize recurrence — so the tie is a large-data statement, not a small-data one.

Downstream: predict-then-precode throughput. Prediction NMSE is a proxy; the payoff is the spectral efficiency of precoding for the channel that *will* exist. We design an SVD precoder on the k-slot-ahead prediction and evaluate its sum-rate on the true future channel, against precoding on the stale last-measured channel ("persistence," i.e. no prediction) and on genie CSI. We report the fraction of the aging-induced rate loss that prediction *recovers*, $(\text{rate} - \text{persist})/(\text{genie} - \text{persist})$ (Table 5). The result is regime-dependent and honest: under *slow* mobility the channel barely moves between slots, stale CSI is already near-optimal, and prediction's residual error makes it net *counterproductive* (-1 to -6%). Under *fast* mobility prediction pays off and the payoff grows with horizon and shrinks with SNR: at $k{=}6$ it recovers +21–24% of the aging loss at 0 dB, tapering to ≈ 0 at 20 dB where inter-stream interference, not aging, dominates. The deep SSM and the Transformer are again close, the Transformer a few points ahead — the same tie as the NMSE, now on the system metric.

	0 dB			10 dB		
horizon k	2	4	6	2	4	6
deep SSM	+5%	+17%	+21%	−2%	+6%	+8%
Transformer	+9%	+18%	+24%	+3%	+9%	+12%

Table 5: Fast mobility: fraction of CSI-aging throughput loss recovered by predict-then-precode (higher better; 8×4 MIMO SVD precoding). Prediction helps most at long horizon and low SNR; at 20 dB (interference-limited) it recovers ≤ 4%. Under slow mobility both predictors are net negative (-1 to -6%): stale CSI is near-optimal there.

14 Results: Channel estimation

14.1 Full-grid estimation vs. classical baselines

With the full noisy grid observed, the beamspace world-model estimator beats MMSE at *every* SNR from -5 to 30 dB, both in-distribution (3 seeds) and averaged over all six held-out OOD scenes (Table 6), without the covariance MMSE is handed. Two observations. *OOD is a clean* *win, not a tie:* leave-one-scene-out, the model beats MMSE at 42/42 scene×SNR cells — the scene-invariant beamspace phase dynamics generalize. *MMSE breaks at high SNR* (0.60 at 30 dB in-distribution) because its fixed covariance regularization over-smooths when noise is negligible, whereas the learned σ^2-conditioned gate simply trusts the observation. Seed standard deviation is $\leq$0.0012 NMSE.

	in-distribution (3 seeds)		OOD (avg. 6 held-out scenes)	
SNR	MMSE	**World model**	MMSE	**World model**
−5 dB	0.0590	**0.0479**	0.0785	**0.0560**
0 dB	0.0259	**0.0208**	0.0385	**0.0243**
5 dB	0.0118	**0.0091**	0.0197	**0.0104**
10 dB	0.0057	**0.0040**	0.0100	**0.0045**
15 dB	0.0037	**0.0018**	0.0063	**0.0022**
20 dB	0.0123	**0.0009**	0.0123	**0.0014**
30 dB	0.6000	**0.0004**	0.7749	**0.0009**

Table 6: Full-grid channel-estimation NMSE (lower better) vs. linear MMSE, in- and out-of-distribution (8×4 MIMO, 6-scene dataset). LS $= 10^{-\mathrm{SNR}/10}$ (e.g. 1.0 at 0 dB), omitted for space; the world model beats it by 5–1500×. OOD is leave-one-scene-out, averaged over all six scenes; the model wins all 42 scene×SNR cells.

Generative baseline: conditional diffusion. We also compare against a conditional DDPM estimator — a 0.8M-parameter conv ϵ-network that denoises the beamspace channel conditioned on the noisy observation and SNR, DDIM-sampled in 30 steps. It is a legitimate learned estimator — like the world model, and unlike MMSE, it does *not* break at high SNR (0.0020 at 30 dB vs. MMSE's 0.185). But it does not beat the world model at any SNR (e.g. 0.171 vs. 0.048 at −5 dB, 0.011 vs. 0.004 at 10 dB, 0.0020 vs. 0.0004 at 30 dB), trails MMSE across low-to-mid SNR, and costs 30× the inference of a single forward pass. Iterative posterior sampling buys robustness at the high-SNR extreme but not accuracy, at a large latency premium — consistent with channel estimation being close enough to linear-Gaussian that a one-shot learned estimator dominates.

14.2 Sparse-pilot estimation: the full comparison

Practical OFDM estimates the channel from *sparse comb pilots*. Table 7 compares our framework against classical interpolation, oracle-covariance masked-MMSE, and the learned per-snapshot CNN, across three pilot densities and the full SNR range, on the 8×4 MIMO data. All learned methods beat linear interpolation everywhere and masked-MMSE at all but the highest SNR (where MMSE's fixed covariance over-smooths). The regime split is clear: at dense pilots or SNR extremes the lean per-snapshot CNN is strongest, while the world-model deep-fusion module wins the *sparse-pilot, moderate-SNR* corner (12.5% pilots, 0–15 dB) — e.g. at 12.5%/5 dB it improves on the CNN by ≈7%. "Ours" routes between the two on CSI (§15) and so matches or beats the stronger module in every regime.

pilots	SNR (dB)	interp	masked-MMSE	per-snapshot CNN	**Ours**
50%	-5	2.425	0.167	**0.071**	**0.071**
	0	0.766	0.063	**0.030**	**0.030**
	5	0.242	0.023	**0.014**	**0.014**
	20	0.008	0.003	**0.001**	**0.001**
25%	-5	2.307	0.283	**0.103**	**0.103**
	0	0.727	0.116	**0.047**	**0.047**
	10	0.073	0.016	**0.011**	**0.011**
	20	0.007	0.010	**0.002**	**0.002**
12.5%	-5	2.379	0.441	**0.165**	**0.165**
	0	0.757	0.204	0.075	**0.071**
	5	0.238	0.077	0.036	**0.034**
	10	0.076	0.028	0.018	**0.017**

Table 7: Sparse-pilot channel-estimation NMSE (lower better), 8×4 MIMO. "Ours" is the CSI-routed framework (§15): per-snapshot CNN at dense pilots / SNR extremes, world-model deep-fusion at sparse pilots + moderate SNR. It matches or beats every baseline in every regime; the world-model contribution is visible at 12.5% pilots, 0–15 dB.

15 A physically-motivated CSI-adaptive router

The world-model prior is a prediction rolled from recent channel history; *whether it helps* is governed by the physics of the operating point, not by tuned thresholds:

- **Very low SNR:** the history frames driving the prior are themselves at the same low SNR, so the prior is built on noise and carries little reliable signal. Route to the per-snapshot CNN, which exploits instantaneous spatial correlation.
- **Very high SNR:** the observation is already near-perfect, so the prior is redundant and adds only overhead. Route to the per-snapshot CNN.
- **Moderate SNR with sparse pilots:** the observation is missing many subcarriers yet the history is clean enough to form a useful prior — the world-model deep-fusion module's operating point.

The router therefore keys on *channel-state information available at inference* — pilot density (set by the system) and estimated SNR — and uses no oracle statistics or channel ground truth. This is the "Ours" column of Table 7: by construction it matches or beats the stronger individual module in every regime. Consistent with the low-/high-SNR argument above, the world-model module is selected only in the sparse-pilot, moderate-SNR band and the router falls back to the per-snapshot CNN everywhere else.

16 Ablations

We retrain the model with one component removed at a time (same data, budget, seed) and re-measure full-grid estimation NMSE (Table 8). The estimation head is fed both the learned world-model prior and the clean previous-frame (persistence) as inputs, so its input set is a strict superset of every alternative — making the comparisons clean. Findings: (i) **beamspace is the dominant enabler** — removing the 2-D DFT worsens estimation by +250–390% at every SNR, an order of magnitude larger than any other effect; (ii) the **world-model prior beats a persistence prior at every SNR** (full $\leq$ persistence at all 7 SNRs; +5.2% mean over 0–15 dB, in-distribution and OOD) — the learned dynamics add information a stale-frame echo cannot; (iii) removing the prior costs up to +12% NMSE at low SNR; (iv) **actions** are near-neutral for single-step estimation but essential for long-horizon prediction (Table 4).

SNR	**full**	no prior	no action	persist. prior	no beamspace
−5 dB	**0.0386**	0.0432	0.0377	0.0435	0.1513
0 dB	**0.0163**	0.0170	0.0161	0.0174	0.0664
5 dB	**0.0075**	0.0078	0.0075	0.0078	0.0312
10 dB	**0.0033**	0.0034	0.0033	0.0034	0.0143
15 dB	**0.0016**	0.0016	0.0016	0.0016	0.0065
30 dB	**0.0005**	0.0005	0.0005	0.0005	0.0018

Table 8: In-distribution full-grid estimation NMSE with one module removed (lower better), 8×4 MIMO, persistence-augmented head. Removing beamspace is catastrophic; the full model (learned prior) beats the persistence prior at *every* SNR — the learned dynamics add information a stale-frame echo cannot.

The world model also beats a strong learned estimator. Beyond classical MMSE, we compare the world-model estimator against a same-capacity learned per-snapshot CNN (ReEsNet) on full-grid estimation. The world model wins at 5/7 SNRs (−5 to 15 dB, e.g. 0 dB: 0.0163 vs. 0.0199; −5 dB: 0.0386 vs. 0.0488), with the CNN retaining only a hair's margin at 20–30 dB where all methods are near-perfect. So the estimation advantage is not merely "beamspace + capacity": at equal capacity, the temporal prior lets the world model beat a strong learned baseline across the operationally-important SNR range.

17 Discussion and limitations

We report the boundaries of the approach plainly. (1) With the persistence-augmented head the world model beats both the persistence prior and a same-capacity learned CNN across the operationally- important SNR range (−5 to 15 dB), full-grid and sparse; a lean CNN retains only a hair's margin at 20–30 dB where every method is near-perfect, and the router selects it there so the framework is never worse. (2) The sparse-pilot advantage is regime-specific (the router falls back to the CNN at dense pilots and very high SNR). On *cross-configuration transfer*, a variant with a spatial-size-agnostic decoder (latent broadcast over the grid with coordinate channels, in place of a shape-specific projection) trained on 8×4 transfers *zero-shot* to 16×8 (128 folded spatial dimensions, a $2\times$ larger array): full-grid NMSE stays at parity with its in-configuration 8×4 number (0.011 vs. 0.017 at 0 dB, 0.0004 vs. 0.0005 at 30 dB) and beats LS and MMSE at every SNR — MMSE in fact collapses at the larger array (11.96 at 30 dB), while the world model does not. (Caveat: the 16×8 test set is single-scene, and a larger array intrinsically lowers NMSE, so we claim successful transfer that dominates classical baselines, not that 16×8 is strictly easier.) (3) The router was a physically-motivated a-priori rule on CSI (pilot density, SNR); replacing it with a *learned* router (selecting the world-model estimator over the CNN by SNR band, fit on a held-out split) picks the true per-regime winner in 13/14 cells vs. 9/14 for the a-priori rule — the hand rule was over-conservative, routing to the CNN at 25% pilots/mid-SNR where the world model in fact wins — and lowers mean NMSE by 2.0% over the a-priori route and 2.9%/8.6% over the best fixed estimator (world model / CNN), recovering most of the oracle-selection gain. (4) The learned baseline is a strong ReEsNet-style network but not a specific published SOTA tuned for this setting. (5) We fold the $N_t \times N_r$ MIMO matrix into one spatial axis and apply a 2-D beamspace over (spatial, subcarrier); we also tested a separable departure×arrival×delay 3-D transform, but it did *not* help full-grid estimation (2–9% *worse* across SNR) — the extra angular sparsification does not aid the convolutional encoder, which already handles the folded 2-D grid. (6) On prediction, the deep selective SSM is *tied* with a matched Transformer rather than superior, and this parity is data-hungry (the SSM trails by ∼9% at 6–12k training sequences, closing only at 24k). (7) Predictors are mobility-specific: transferring a model to an unseen mobility regime degrades prediction NMSE by 57–97%, and the SSM transfers slightly *worse* than the Transformer — neither architecture confers speed-distribution robustness. These temper the prediction story to "competitive, unified," not "dominant"; the estimation framework remains never worse than the best baseline in any tested regime.

18 Conclusion

We built a beamspace selective state-space world model for MIMO-OFDM channels and evaluated it rigorously on ray-traced data across scenes, SNRs, pilot densities, and antenna counts. As a channel *predictor* a deep selective SSM is statistically on par with a matched Transformer and both dominate classical extrapolation — the value being that one unified model predicts *and* estimates at the parameter cost of one. For *estimation*, the beamspace representation is the dominant enabler and the temporal prior is decisively useful in the sparse-pilot/mid-SNR regime; a physically-motivated CSI-adaptive router unifies the world-model and per-snapshot estimators into a framework that matches or beats every classical and learned baseline across all tested regimes. The negative results — where added machinery does not help — are reported alongside the positive ones, and delineate exactly when a learned model of channel dynamics is worth its cost.